PROFILEBOOK

Capstone Project

Mentored by:

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ABSTRACT

This document provides a detailed overview of the ProfileBook application, an online social media platform that enables users to create posts, send messages, comment, and report inappropriate behavior. Admins manage users, approve posts, and handle reports. The project utilizes Angular, ASP.NET Core MVC, SQL Server, and SignalR for real-time communication.

CHAPTER 1: INTRODUCTION

1.1 Motivation

ProfileBook connects users virtually, enabling easy interaction through posts and messaging while maintaining a secure and moderated environment.

1.2 Problem Statement

Users face fragmented services when it comes to social interaction, requiring multiple apps for messaging,
posting, and reporting. ProfileBook addresses this by offering a unified platform.

1.3 Objective of the Project

- Secure authentication for users and admins.
- Post creation and admin approval.
- Commenting and liking posts.
- Real-time messaging with SignalR
- Reporting system for inappropriate content.
- Image storage and management.

1.4 Theory

The project is designed to provide seamless user experience using Angular for frontend, ASP.NET Core MVC Web API for backend, and SQL Server database. JWT is used for secure authentication, while EF Core manages the database interactions.

CHAPTER 2: PROFILEBOOK

2.1 Features

1. User Authentication (Login, Register, Logout)
2. Post Creation and Approval Process
3. Comment and Like Posts
4. Messaging between Users (SignalR)
5. Reporting UsersSearch for Other Users
6. Admin User and Post Management
7. Group Creation and Management

2.2 Tech Stack

Frontend (Angular): Angular Router for navigation, Axios for API requests, Context API or Redux for state

• Backend: ASP.NET Core MVC with Web API

• Database: SQL Server (Entity Framework Core ORM)

• Image Storage: Server folder

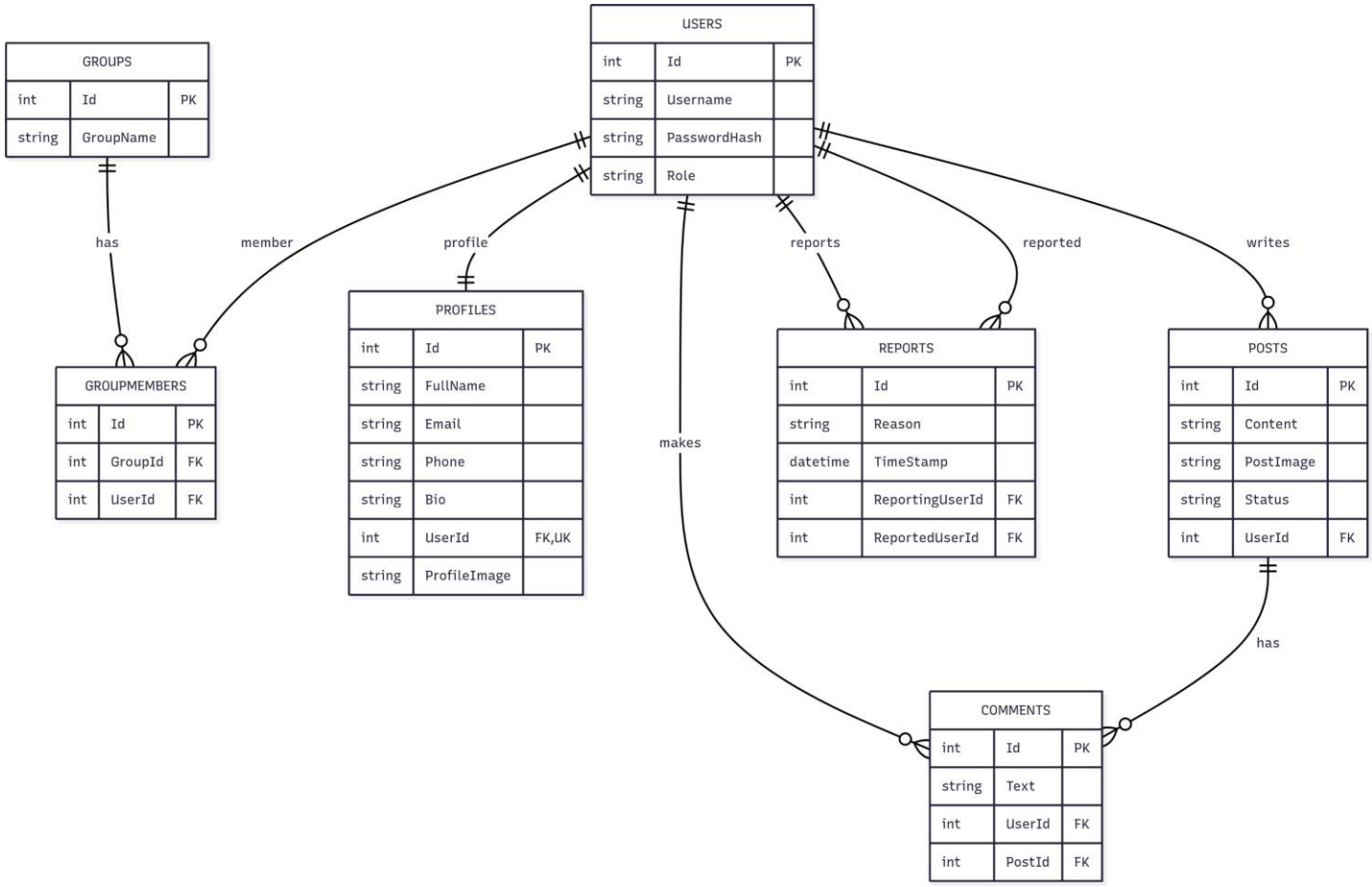
• API Documentation: Swagger

• Real-Time Communication: SignalR (Optional)

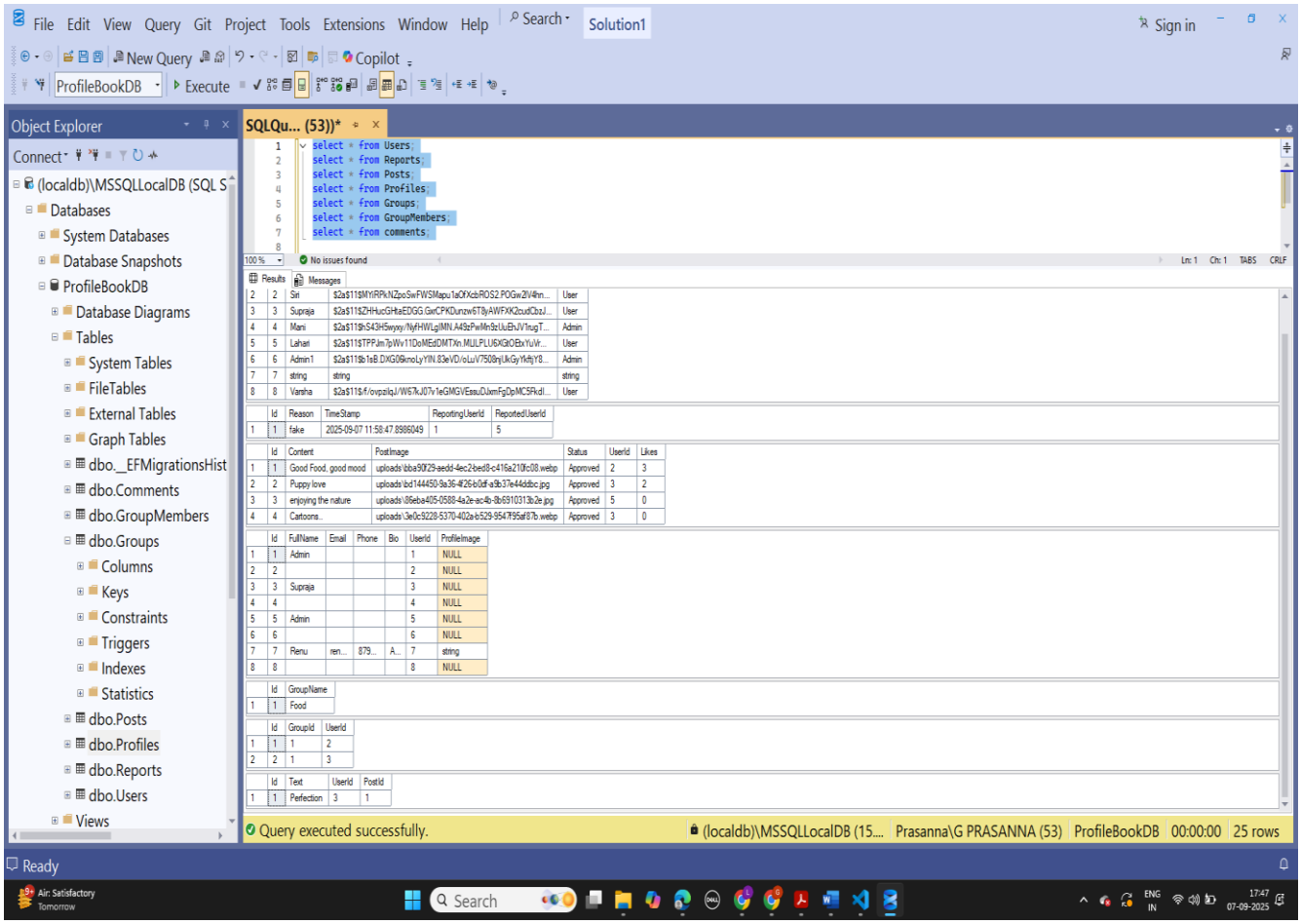
2.3 Architecture Overview

Frontend (Angular): Angular Router for navigation, Axios for API requests, Context API or Redux for state management. Backend (ASP.NET Core MVC): JWT Authentication, EF Core for DB interactions, Web API controllers (UserController, PostController, etc.). Database (SQL Server): Tables - Users, Posts, Comments, Messages, Reports, Groups, GroupMembers, Profiles. Image Upload: Handled via IFormFile, images stored in server directories. Notifications: Optional real-time notifications using SignalR.

2.4 Entity Relationship Diagram (ERD)



Tables


File Edit View Query Git Project Tools Extensions Window Help
Search
Solution1
Sign in
New Query
Copilot
ProfileBookDB
Execute
Object Explorer
Connect
(localdb)\MSSQLLocalDB (SQL S
Databases
System Databases
Database Snapshots
ProfileBookDB
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.__EFMigrationsHist
dbo.Comments
dbo.GroupMembers
dbo.Groups
Columns
Keys
Constraints
Triggers
Indexes
Statistics
dbo.Posts
dbo.Profiles
dbo.Reports
dbo.Users
Views
SQLQu... (53))*
select * from Users;
select * from Reports;
select * from Posts;
select * from Profiles;
select * from Groups;
select * from GroupMembers;
select * from comments;
No issues found
Results
Messages
Query executed successfully.
(localdb)\MSSQLLocalDB (15....
Prasanna\G PRASANNA (53)
ProfileBookDB
00:00:00
25 rows
Ready
Search

CHAPTER 3: REQUIREMENTS AND RESULTS

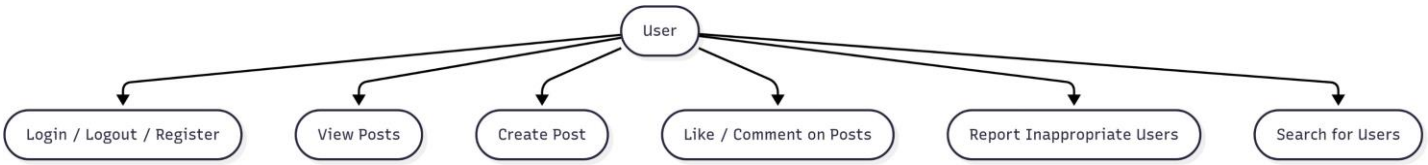
.

3.1 System Requirements

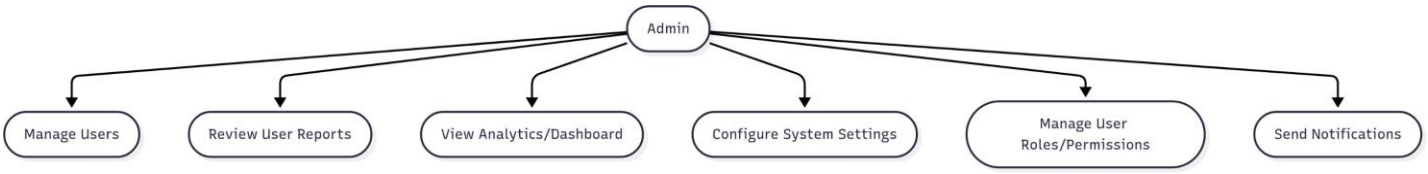
Software: Visual Studio, VS Code, SQL Server, Angular, ASP.NET Core MVC, EF Core. Hardware: Minimum 4 GB RAM, Intel/AMD dual-core, 500 MB free disk space.

3.2 Case Diagrams

User Operations Use Case Diagram



Admin Operations Use Case Diagram



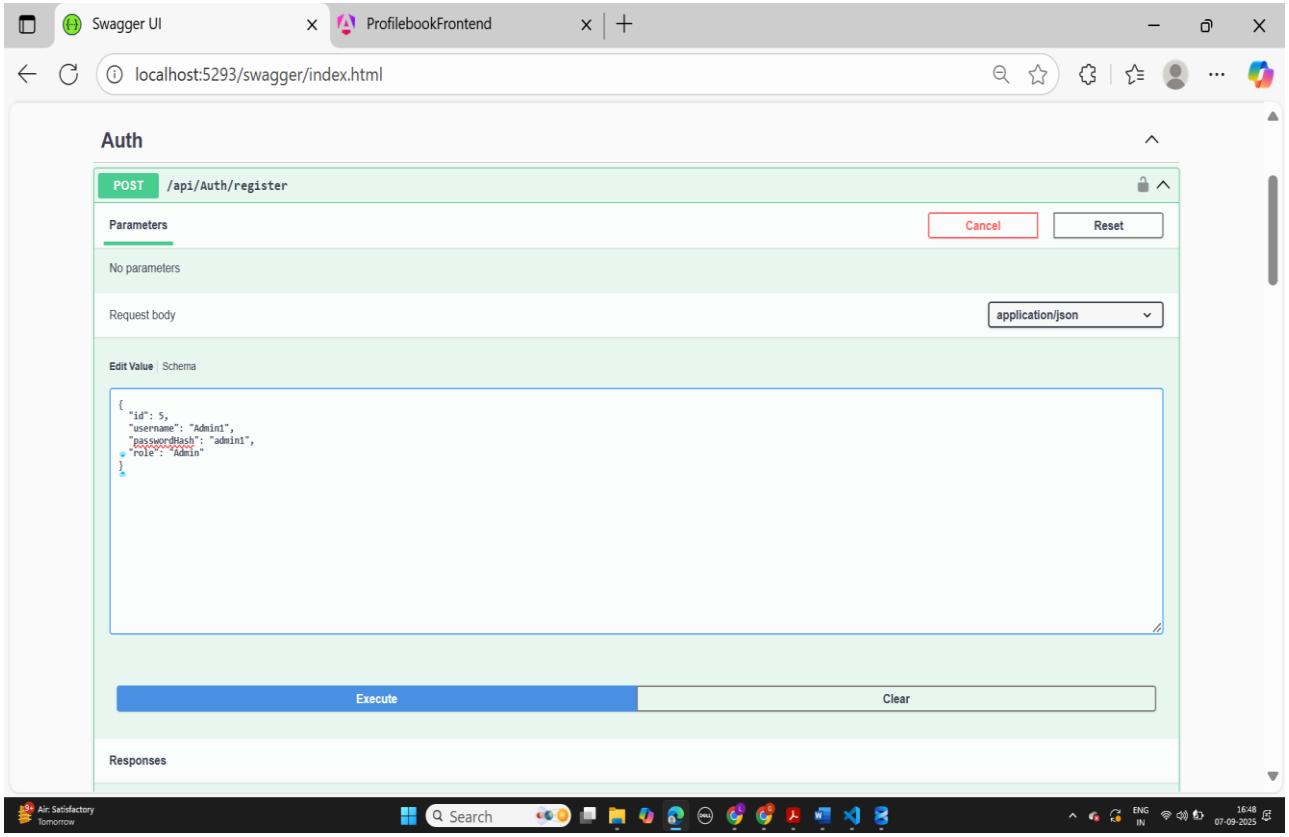
3.3 Implementation

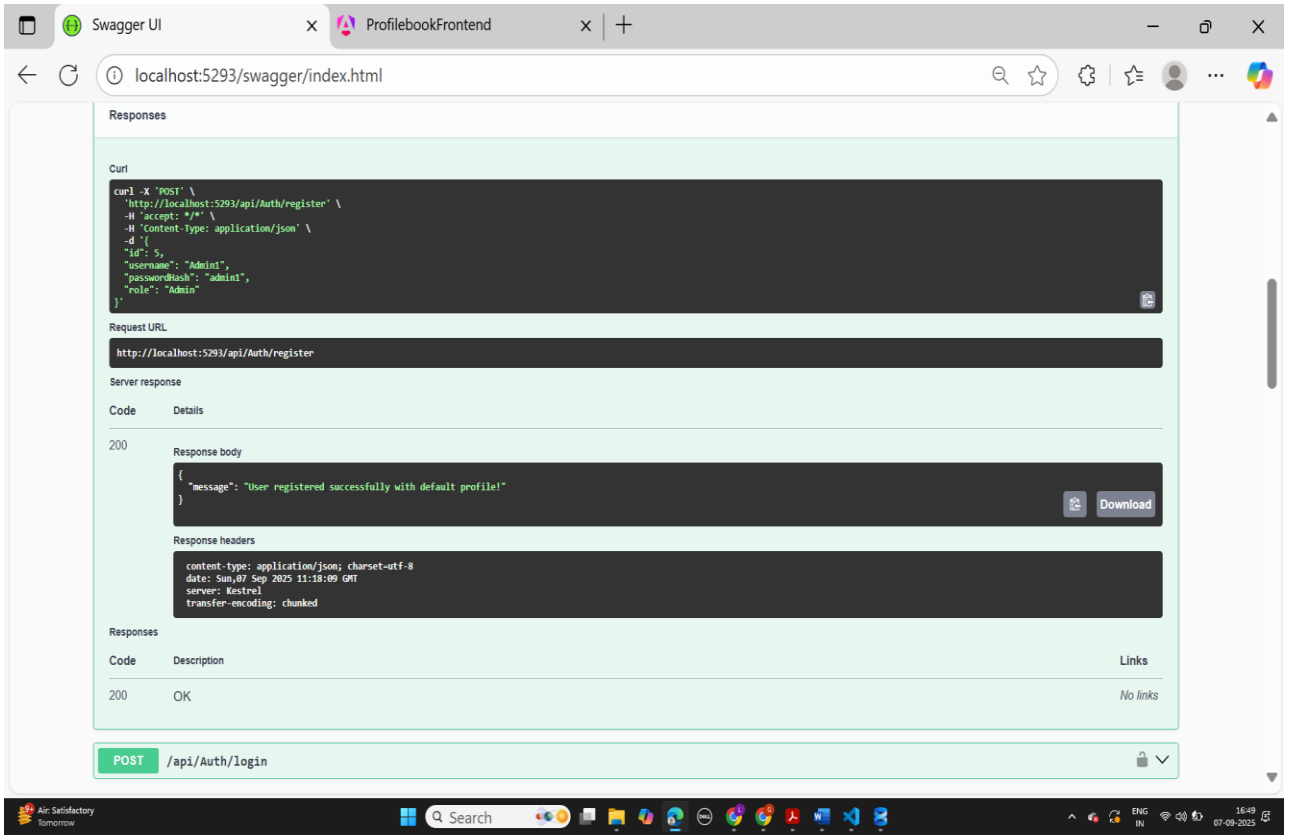
Backend Implementation: Developed Web API endpoints, JWT Authentication, SignalR for messaging, CRUD for Posts, Comments, Users, Reports.

Frontend Implementation :Context API state management.

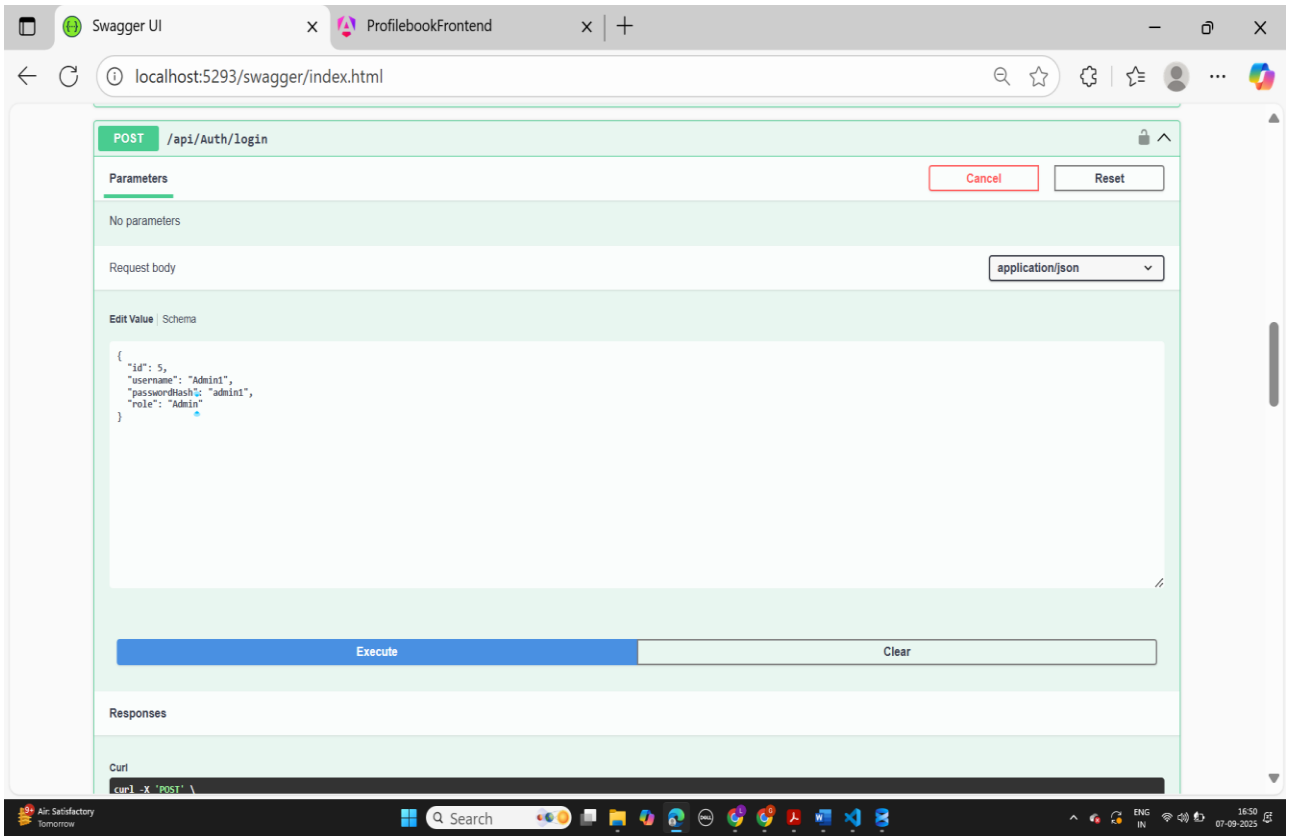
Implementation Screenshot

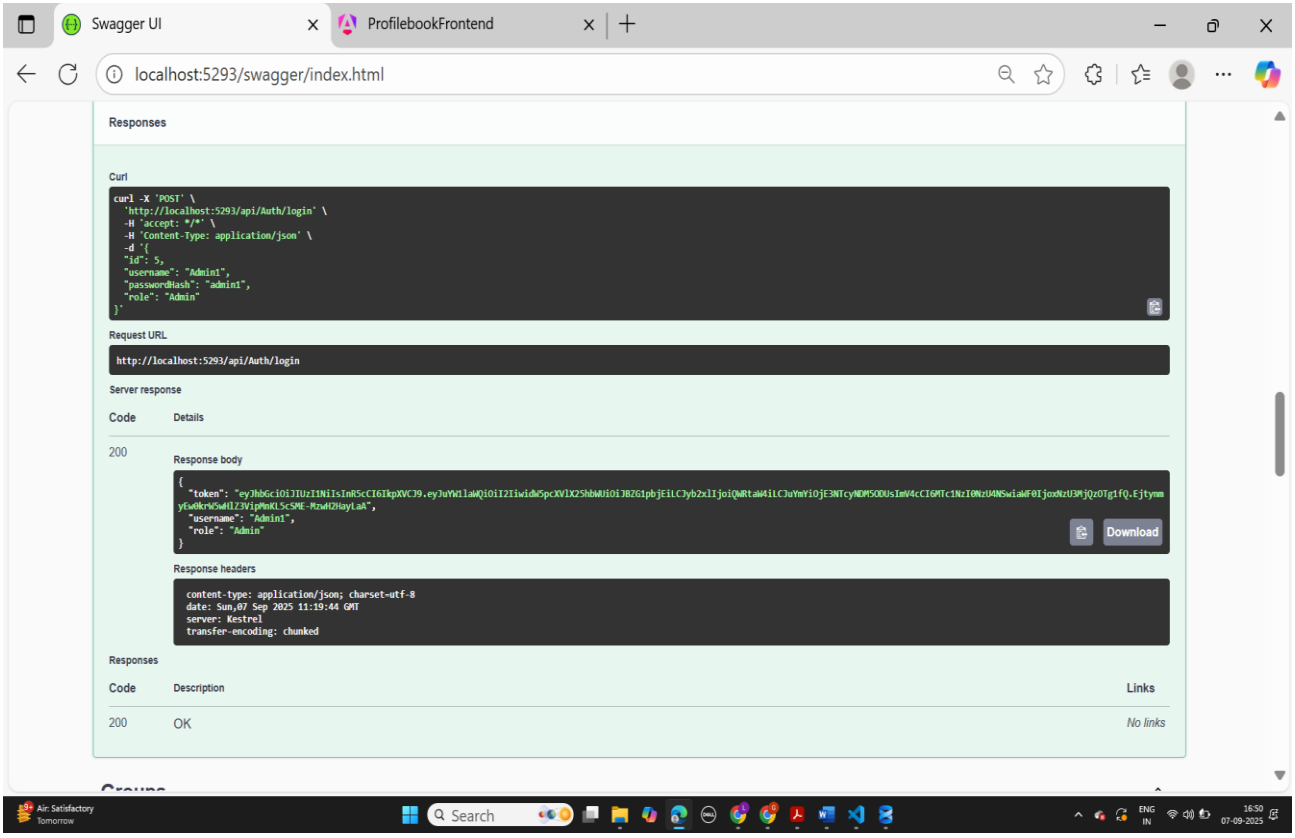
Register API



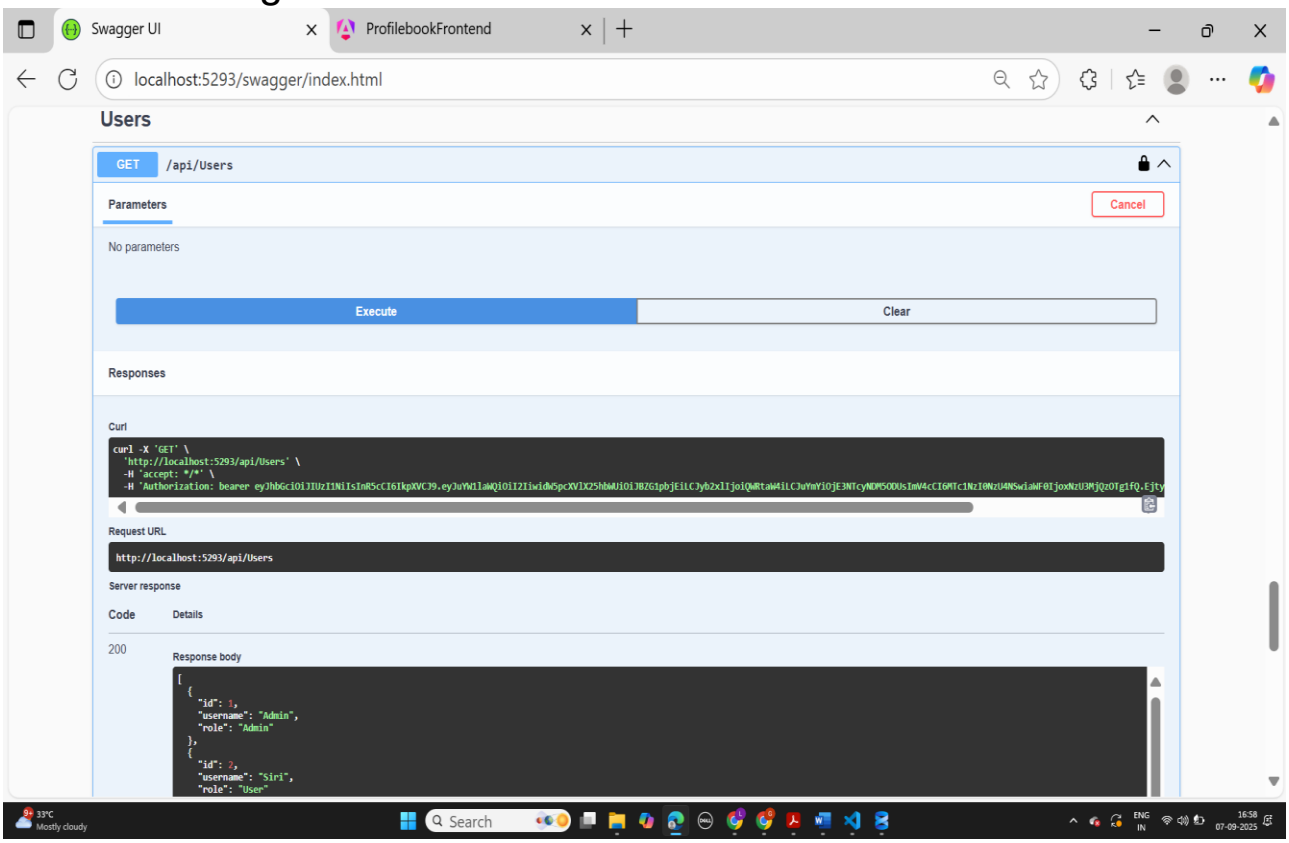


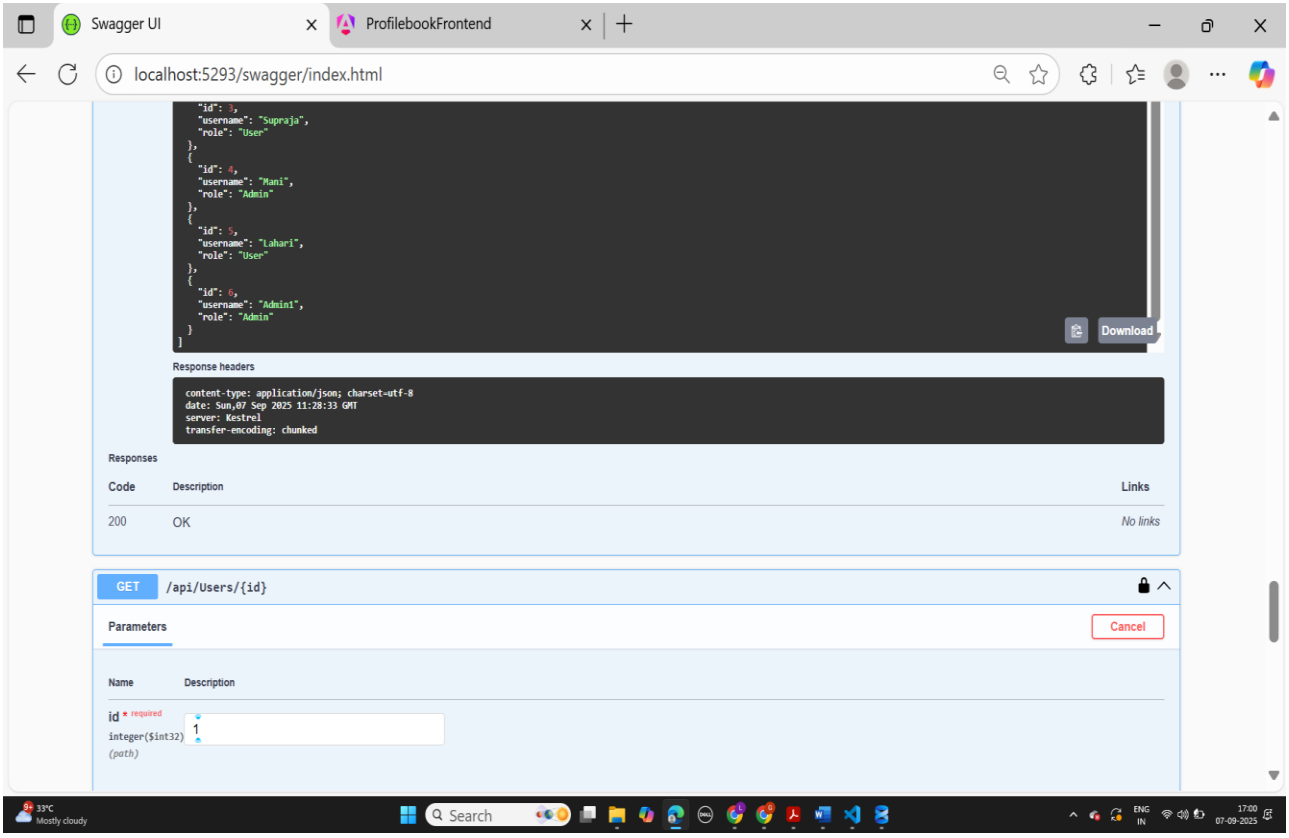
Login API


Swagger UI
ProfilebookFrontend
localhost:5293/swagger/index.html
POST
/api/Auth/login
Parameters
Cancel
Reset
No parameters
Request body
application/json
Edit Value | Schema
{
"id": 5,
"username": "Admin1",
"passwordHash": "admin1",
"role": "Admin"
}
Execute
Clear
Responses
Curl
curl -X 'POST' \

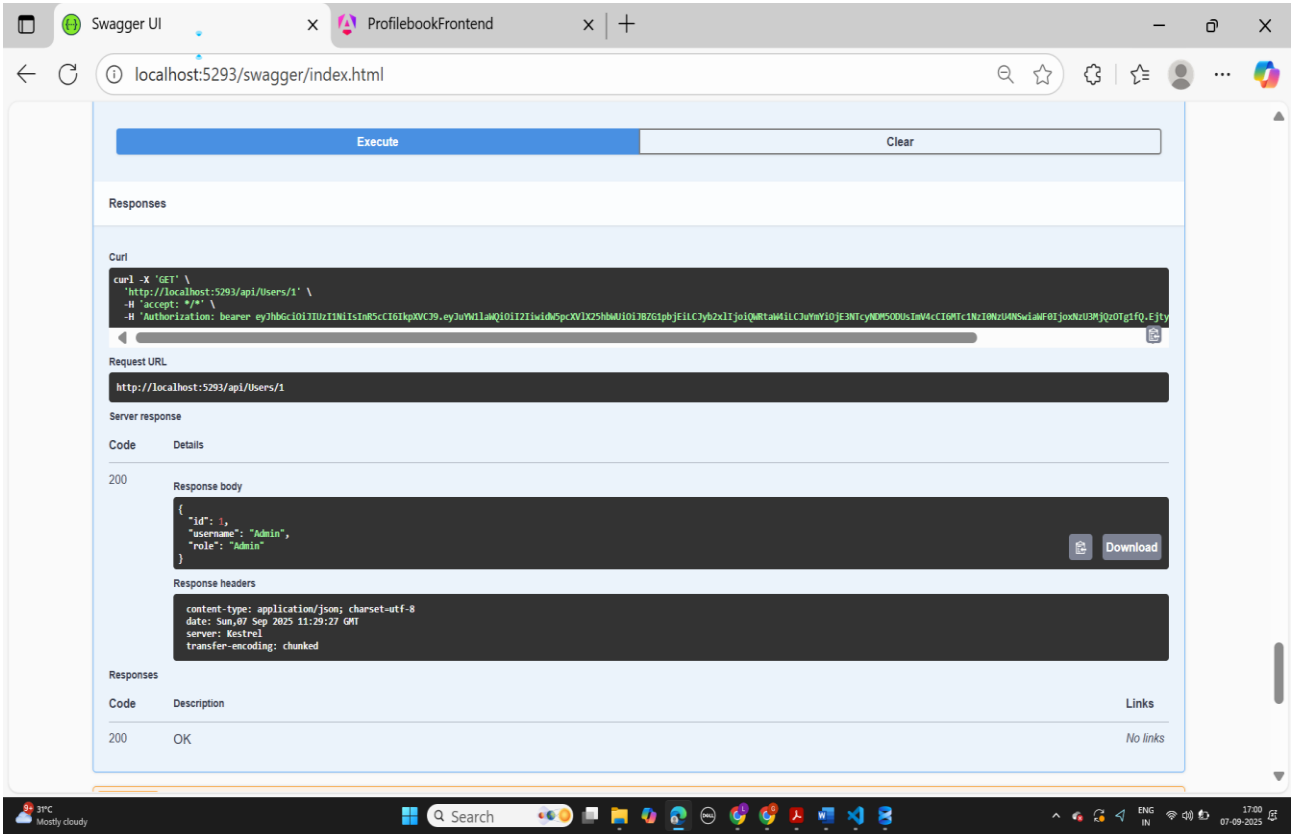

Swagger UI
ProfilebookFrontend
localhost:5293/swagger/index.html
Responses
Curl
curl -X 'POST' \
'http://localhost:5293/api/Auth/login' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
"id": 5,
"username": "Admin1",
"passwordHash": "admin1",
"role": "Admin"
}'
Request URL
http://localhost:5293/api/Auth/login
Server response
Code
Details
200
Response body
{
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJuYW1laWQiOiI2IiwidW5pcXVlX25hbWUiOiJBZG1pbjEiLCJyb2xlIjoiQWRtaW4iLCJuYmYiOjE3NTcyNDM5ODUsImV4cCI6MTc1NzI0NzU4NSwiaWF0IjoxNzU3MjQzOTg1fQ.EjtymnyEwOkrW5wHlI3VipMnKL5cSME-MzwH2HaylaA",
"username": "Admin1",
"role": "Admin"
}
Download
Response headers
content-type: application/json; charset=utf-8
date: Sun,07 Sep 2025 11:19:44 GMT
server: Kestrel
transfer-encoding: chunked
Responses
Code
Description
Links
200
OK
No links

Admin fetching users

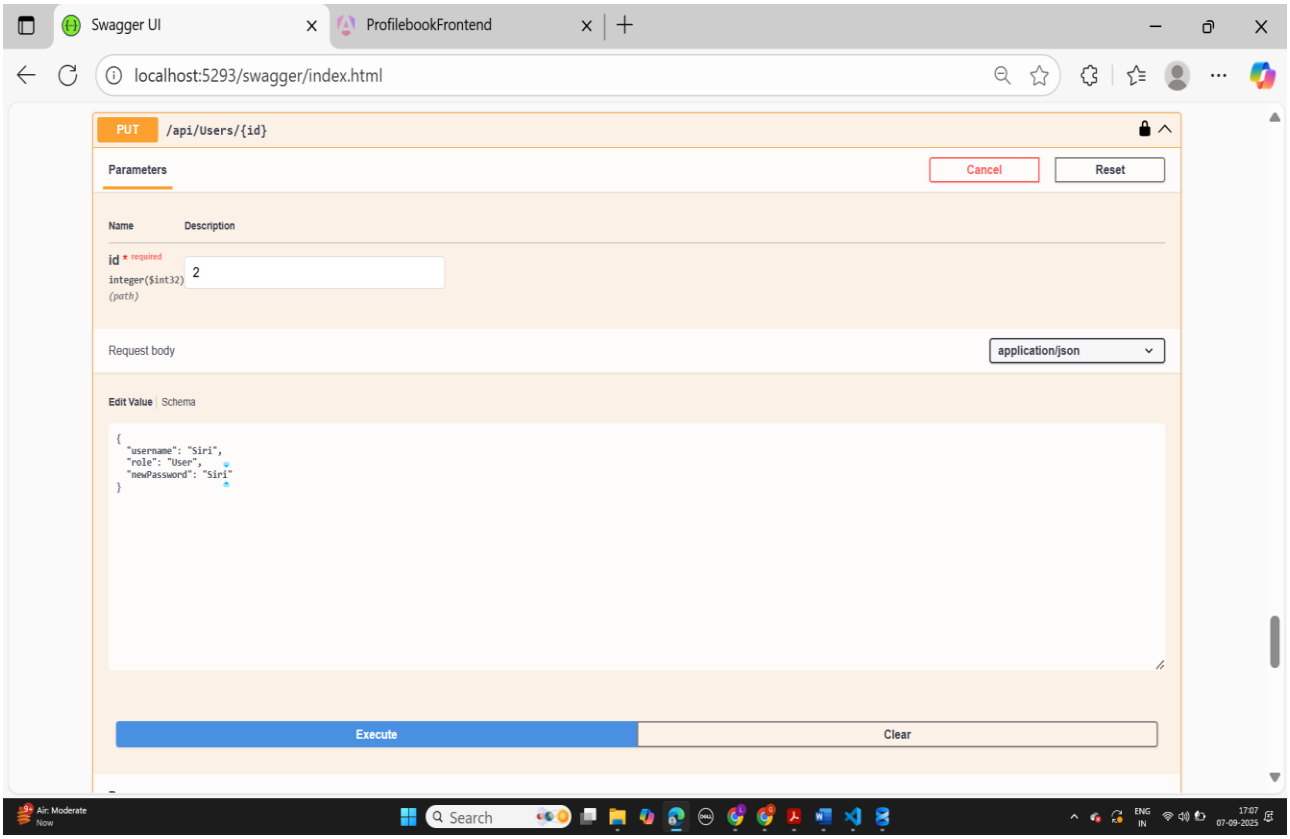


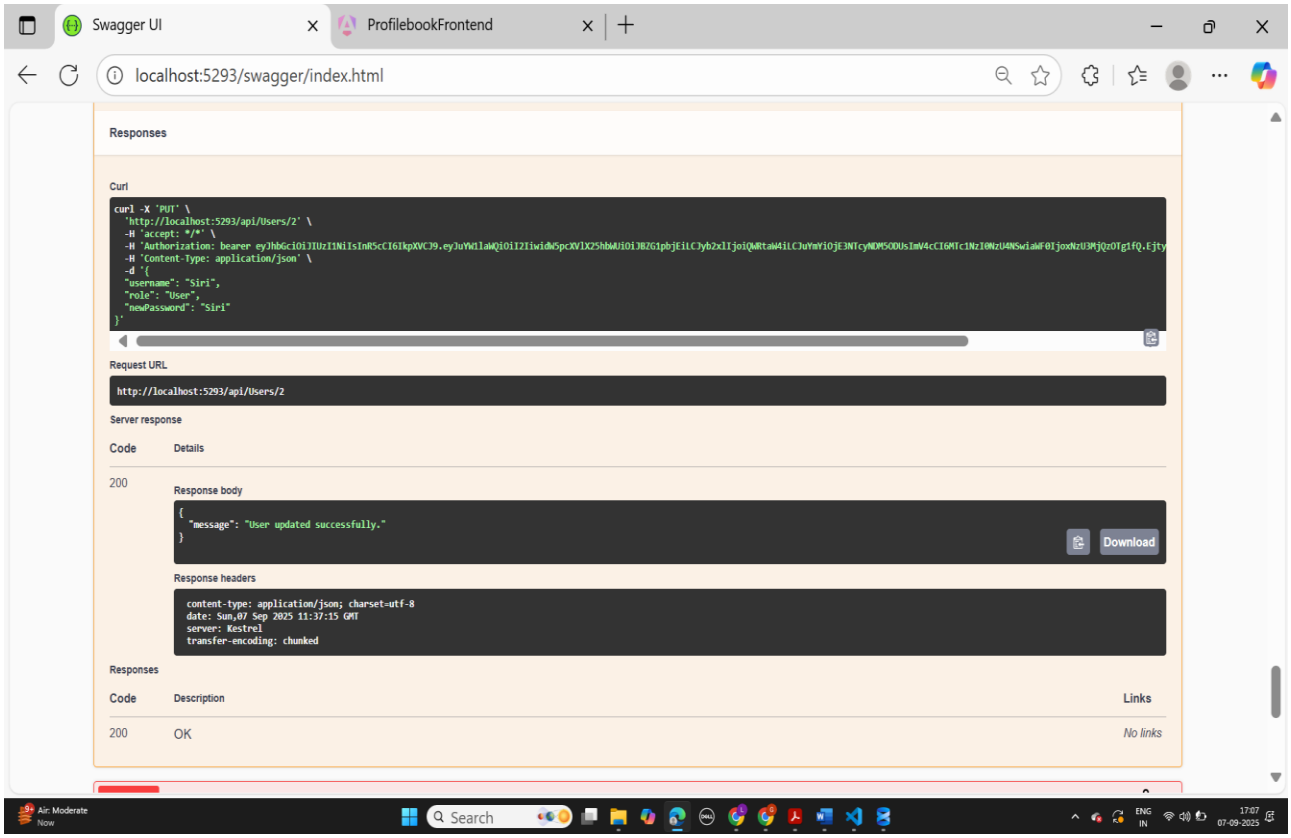


Fetching users by id

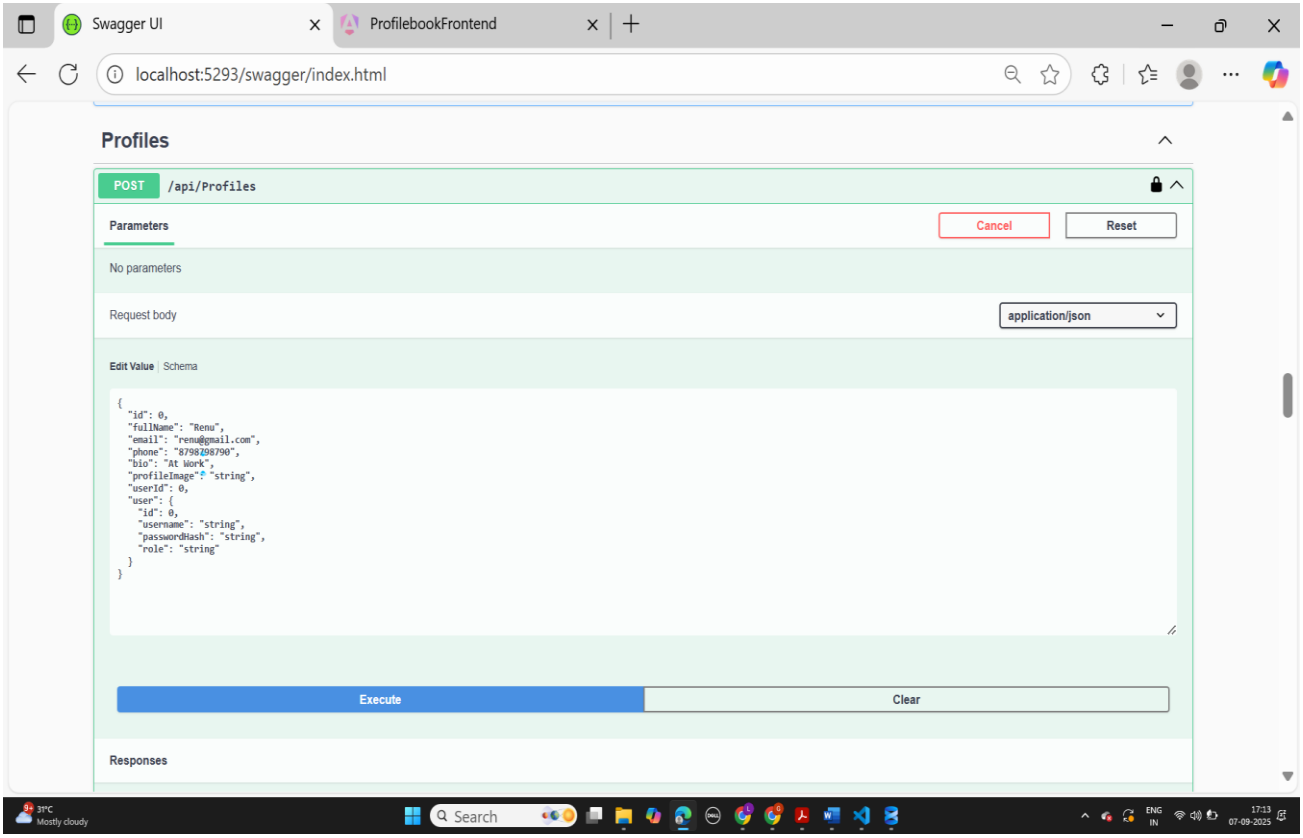


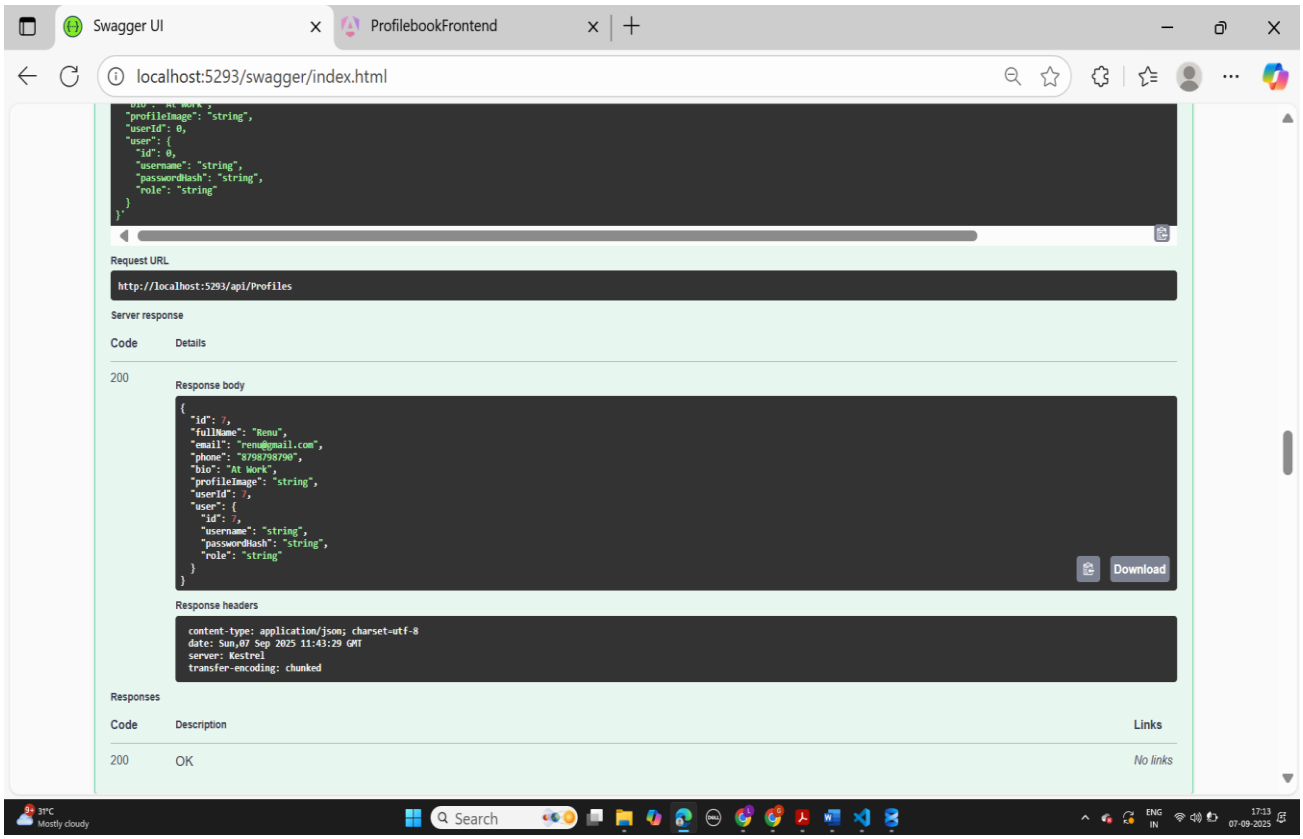
Updating user data





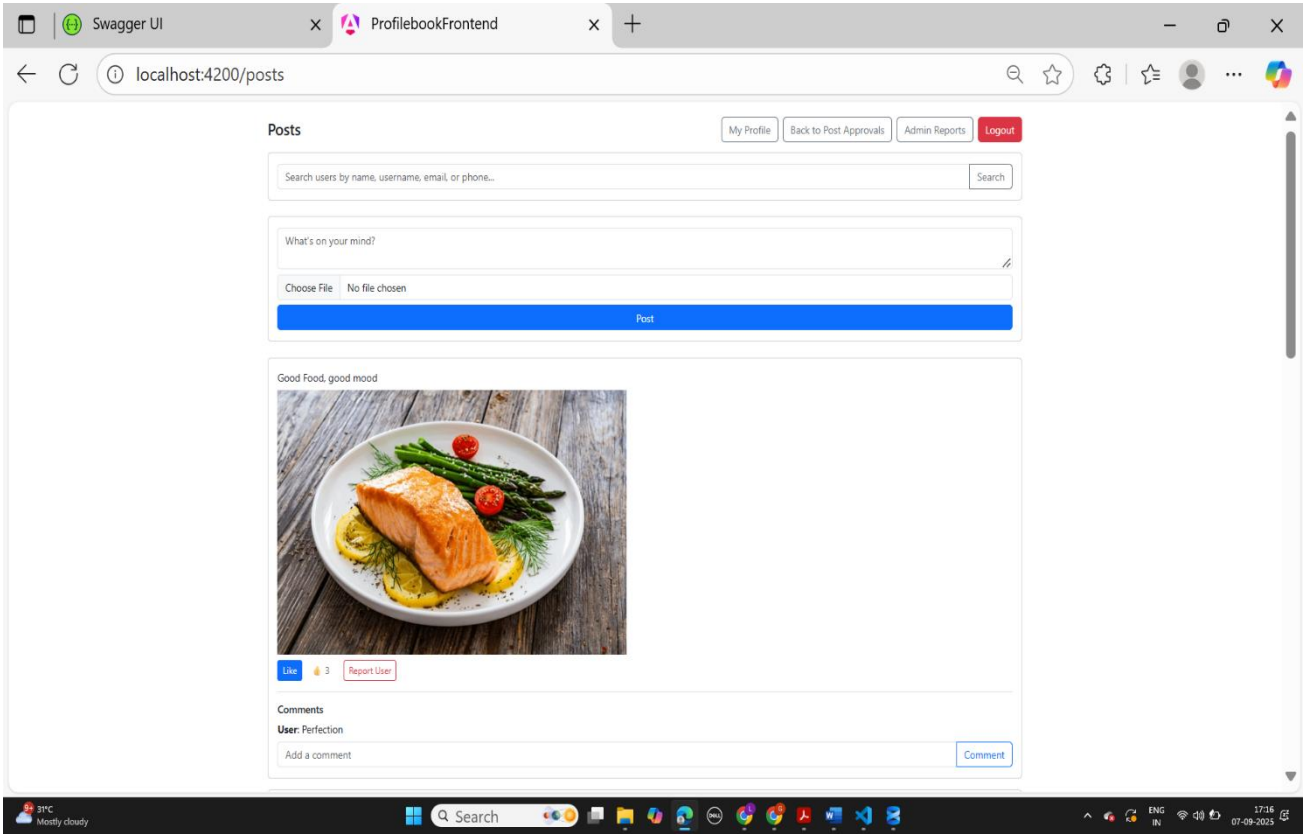
Creating Profile API



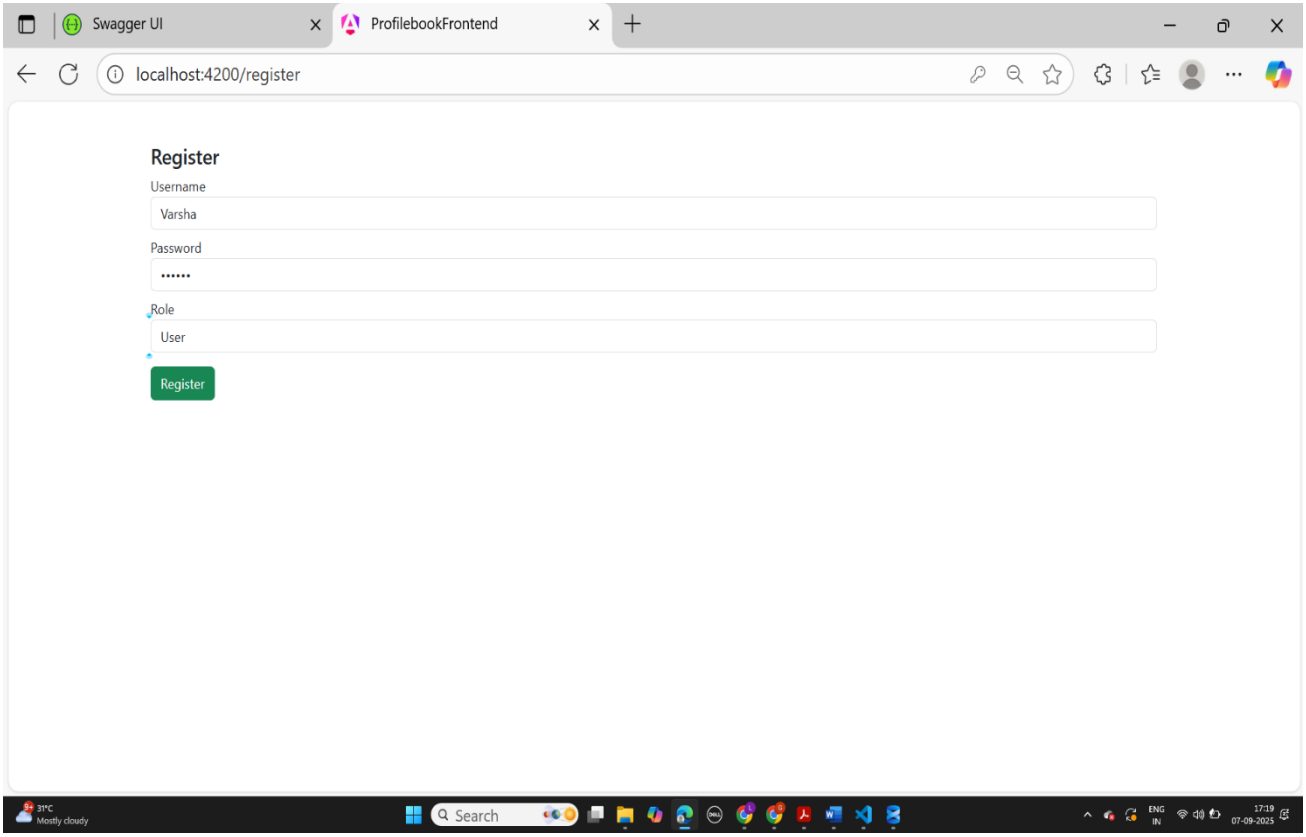

Swagger UI
ProfilebookFrontend
localhost:5293/swagger/index.html
"profileImage": "string",
"userId": 0,
"user": {
"id": 0,
"username": "string",
"passwordHash": "string",
"role": "string"
}
}'
Request URL
http://localhost:5293/api/Profiles
Server response
Code
Details
200
Response body
{
"id": 7,
"fullName": "Renu",
"email": "renu@gmail.com",
"phone": "8798798790",
"bio": "At Work",
"profileImage": "string",
"userId": 7,
"user": {
"id": 7,
"username": "string",
"passwordHash": "string",
"role": "string"
}
}
Download
Response headers
content-type: application/json; charset=utf-8
date: Sun,07 Sep 2025 11:43:29 GMT
server: Kestrel
transfer-encoding: chunked
Responses
Code
Description
Links
200
OK
No links
Search

3.4 Results

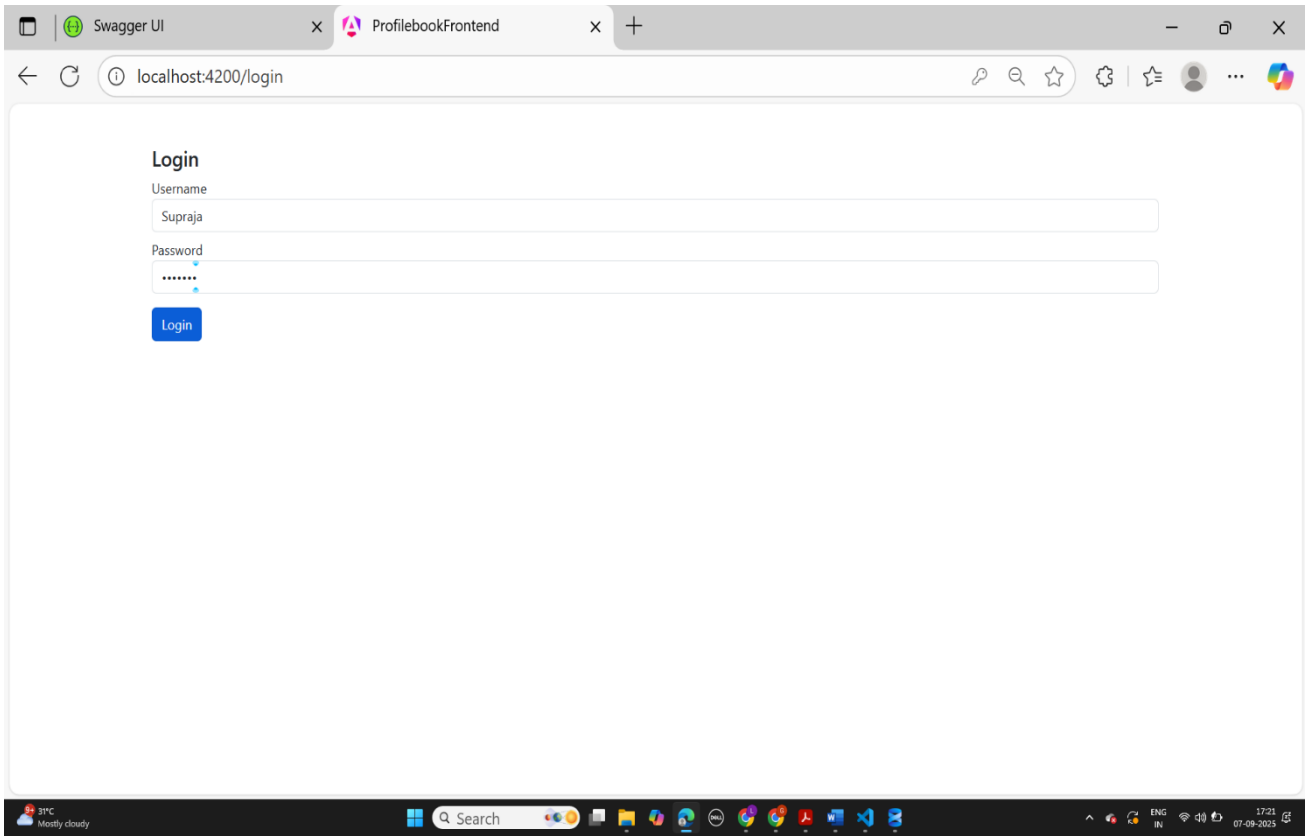
Home Page Displaying Posts:



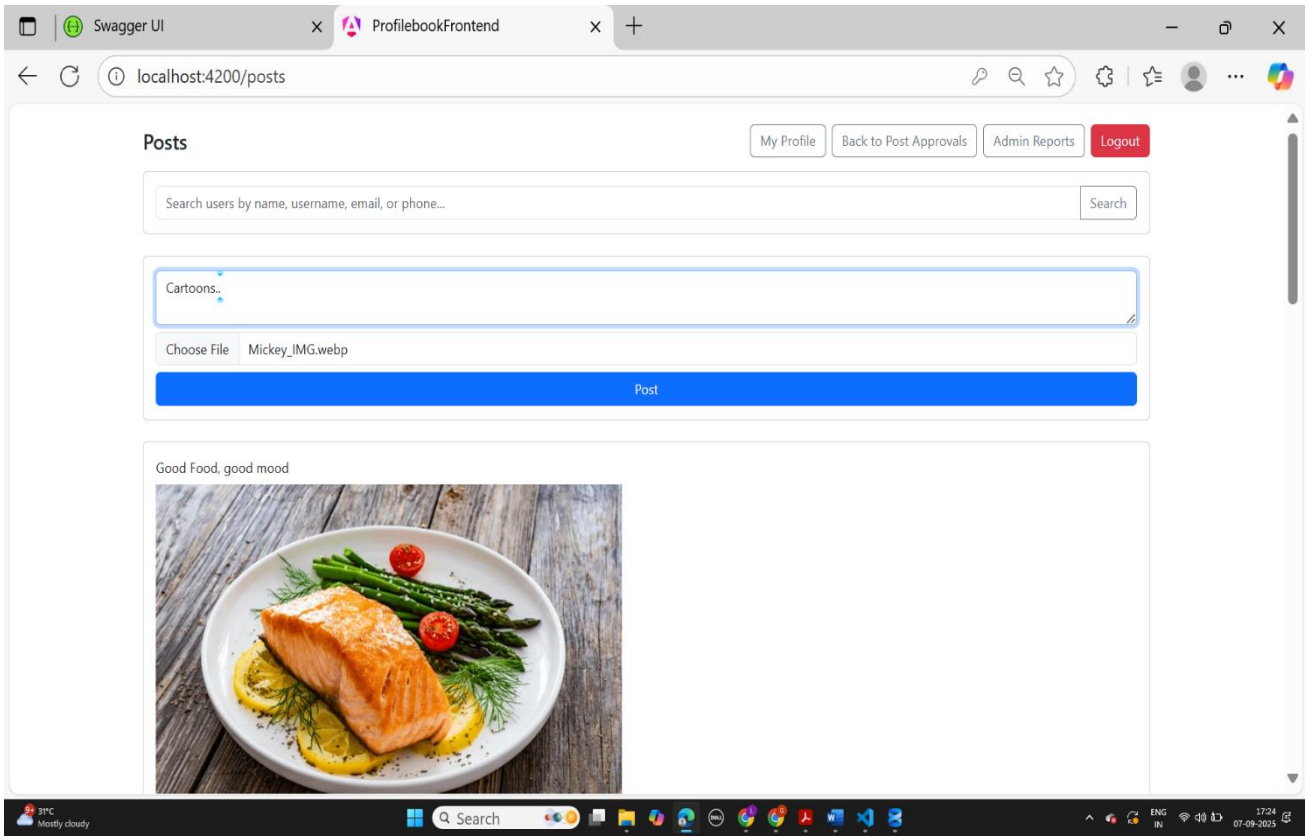
User Registration Page:



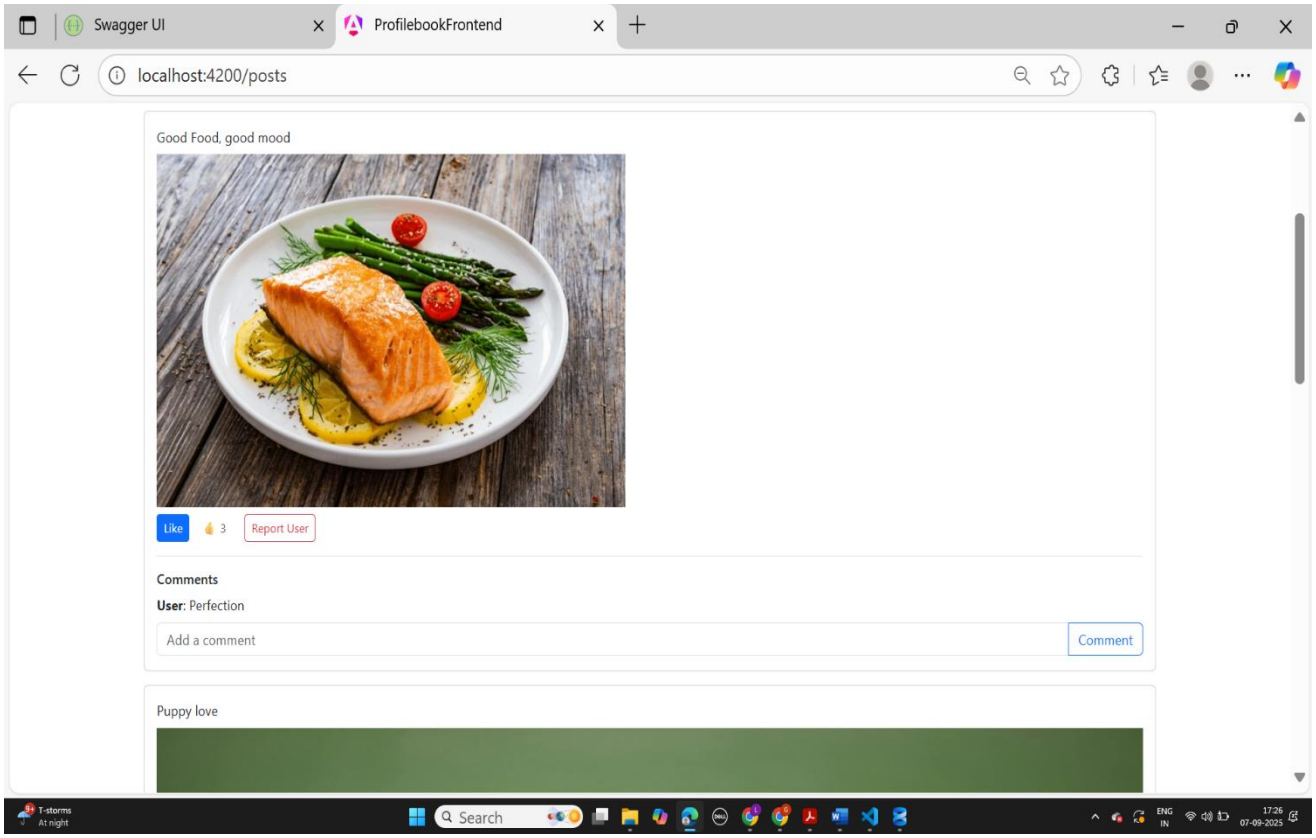
Login Page:



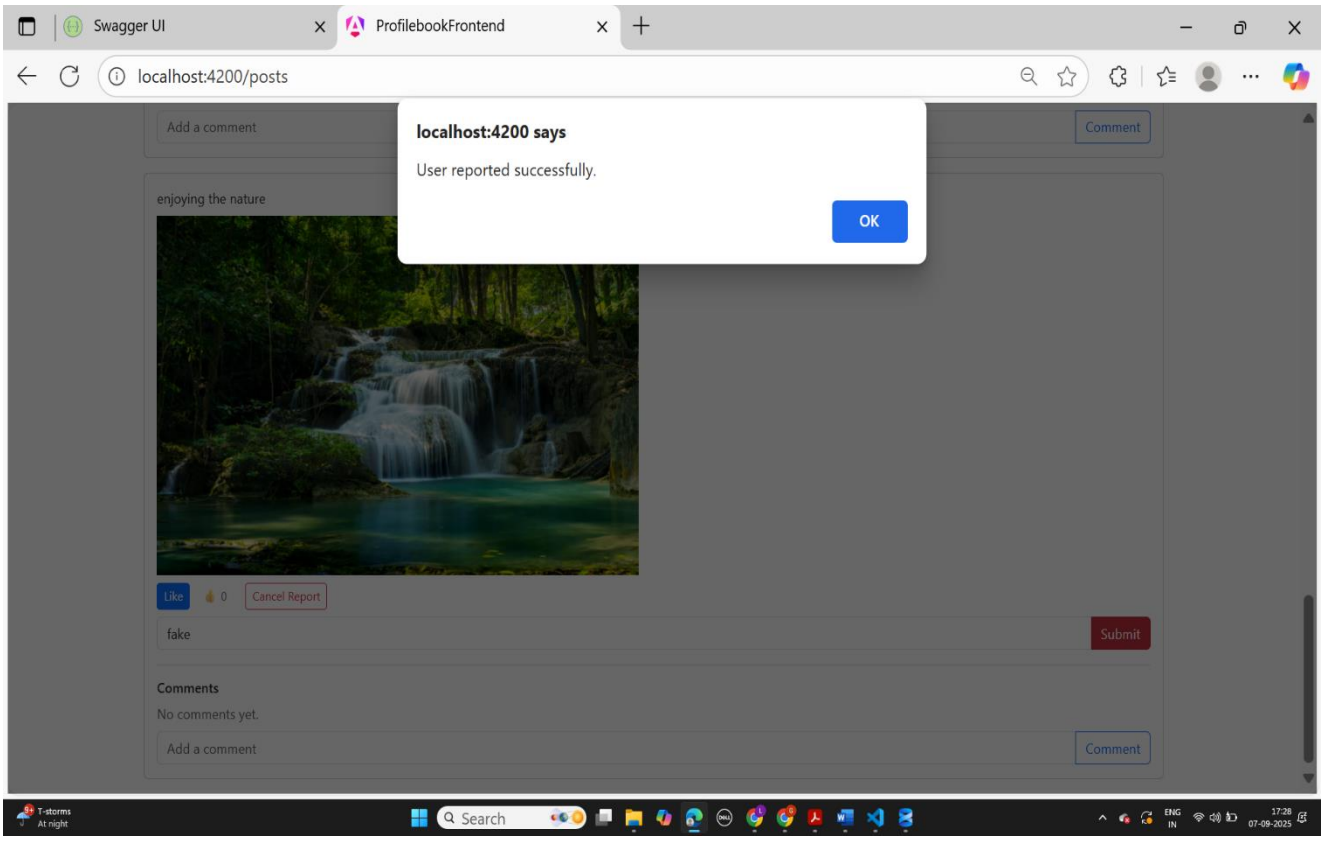
Post Creation Interface:



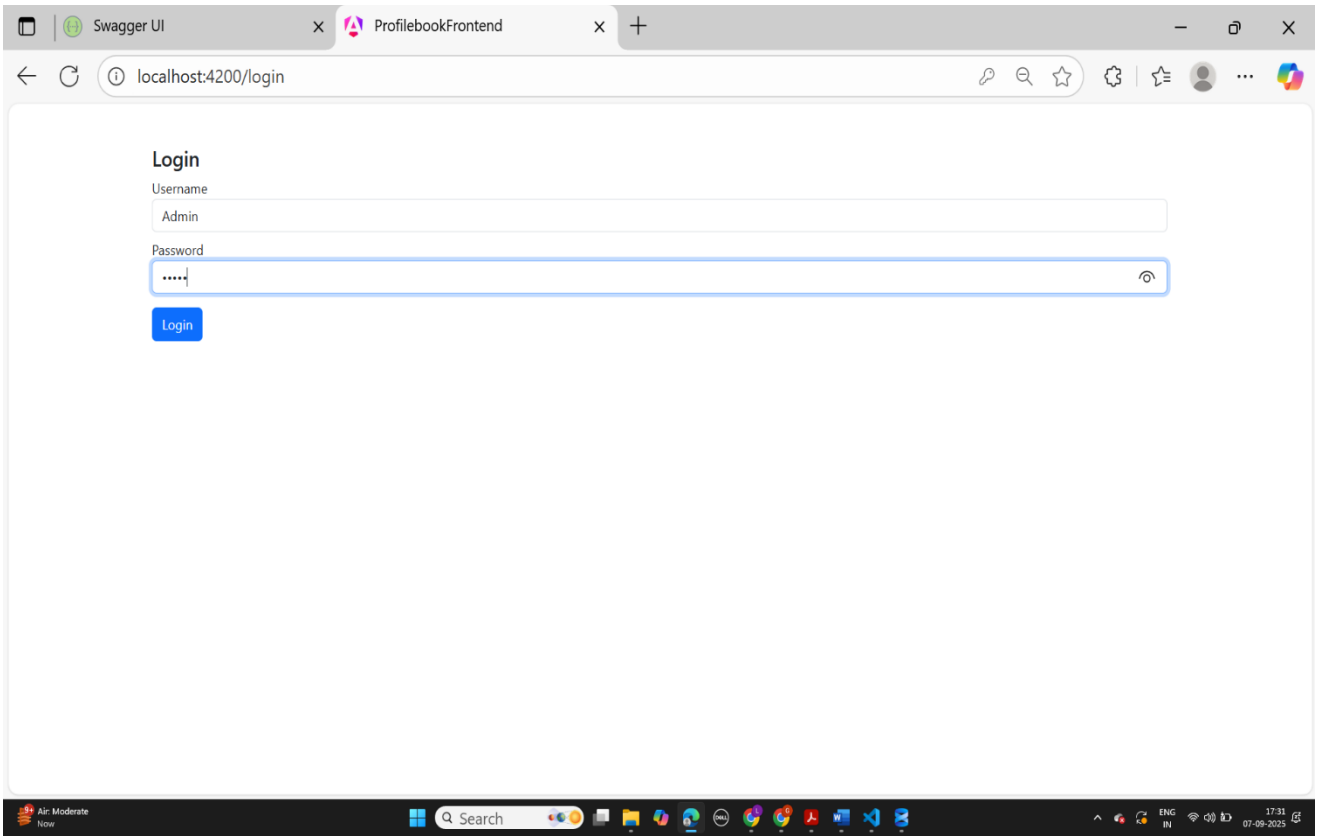
Comment/Message Interface:



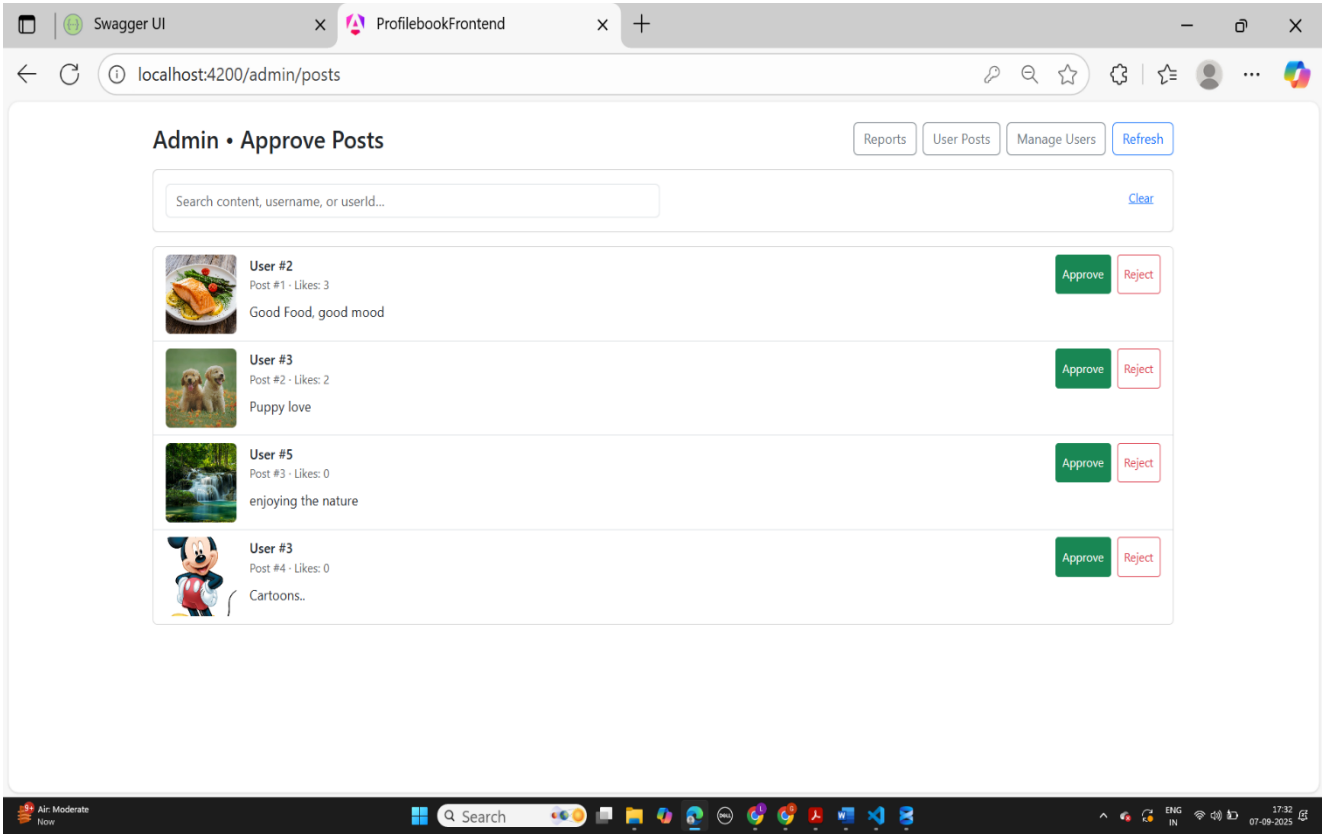
Report User Page:

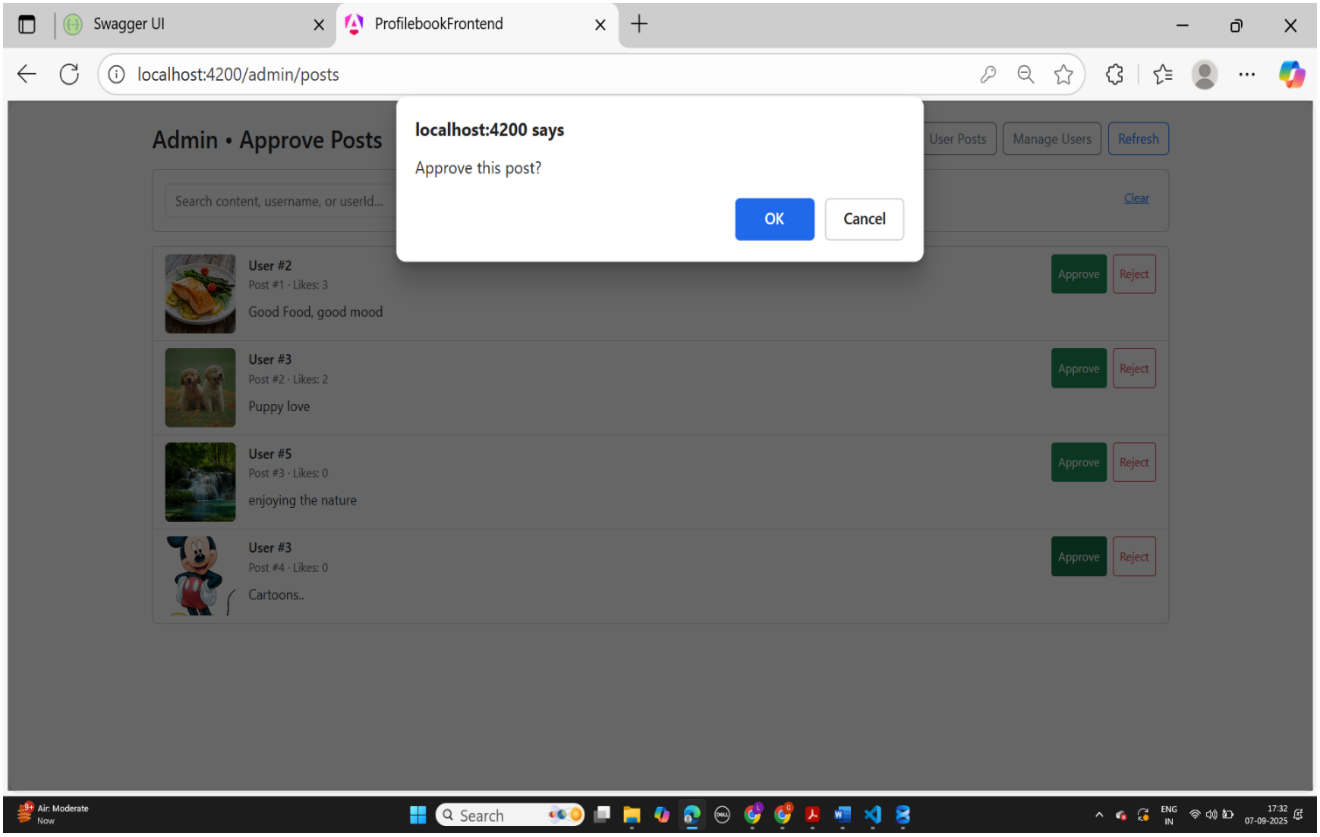


Admin Login Page:

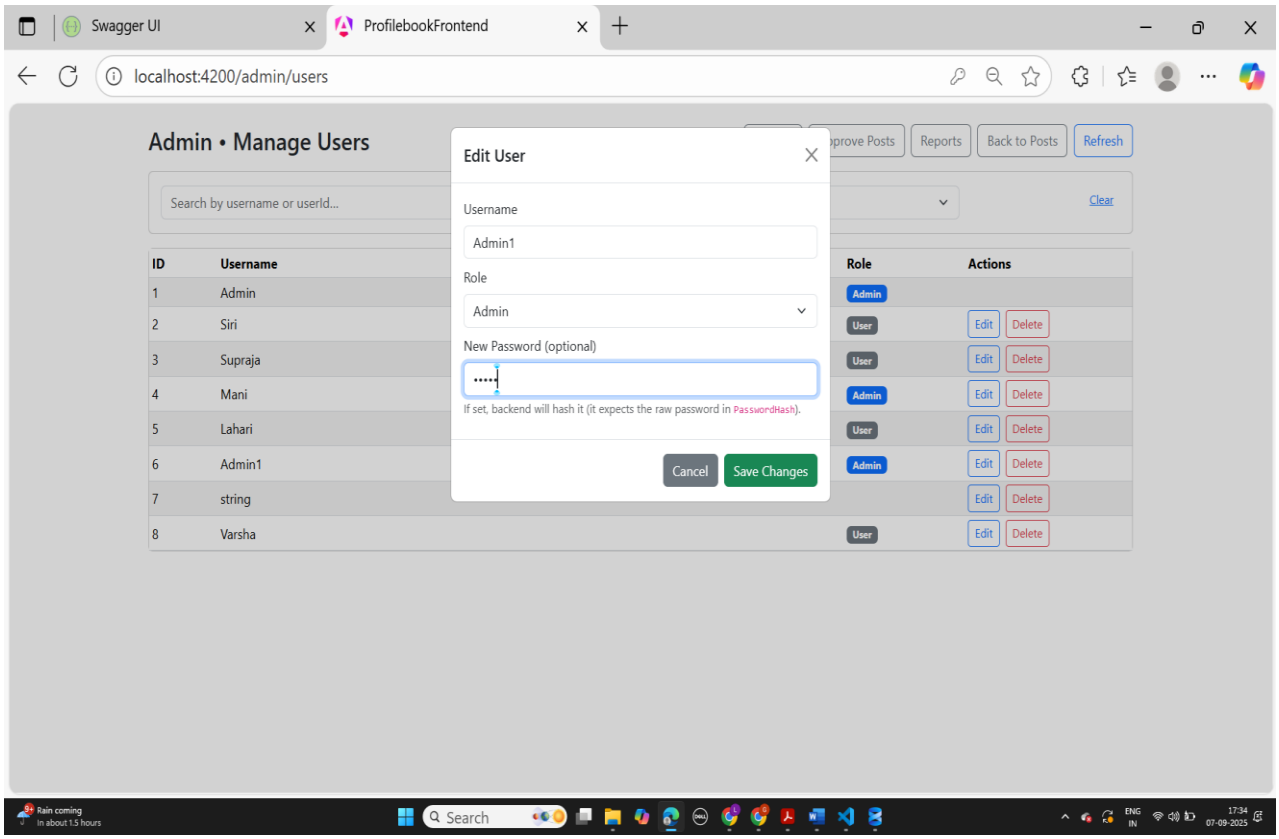


Post Approval Dashboard:

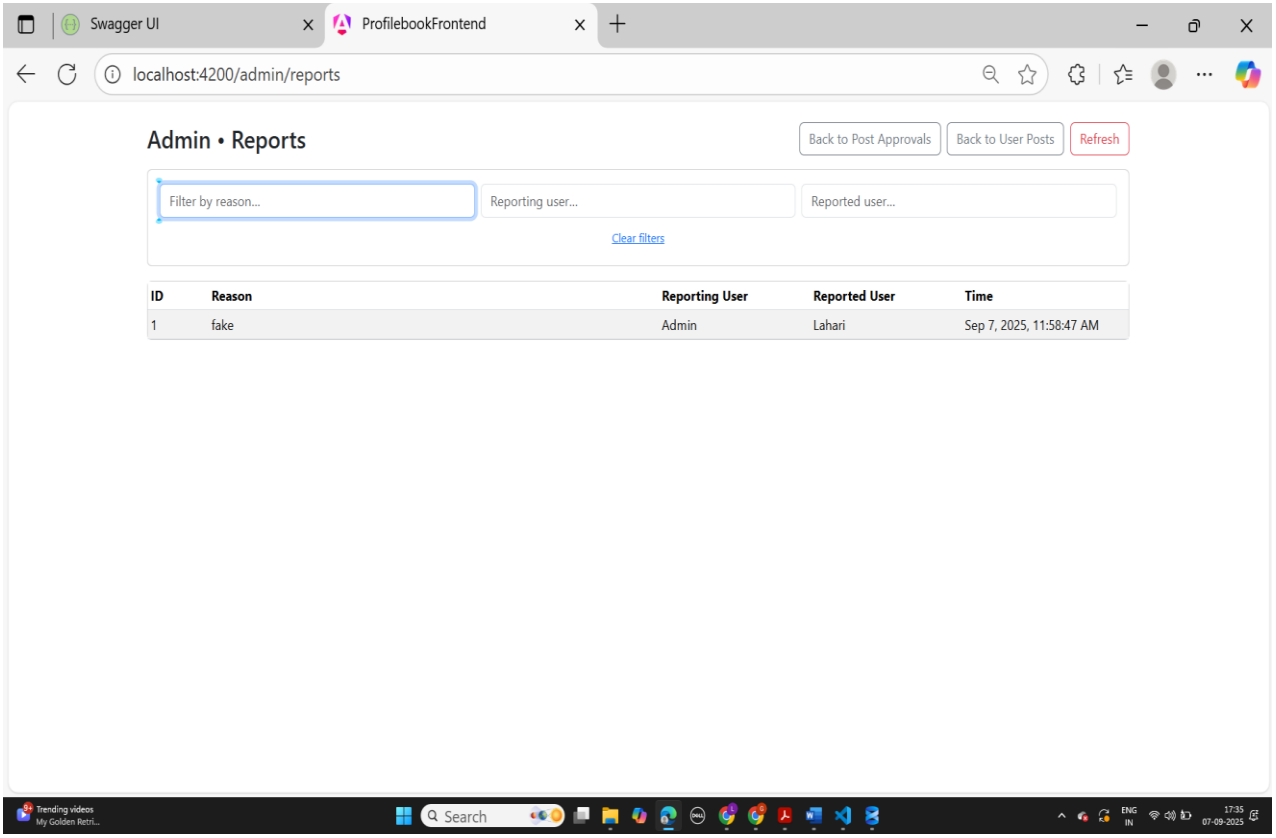




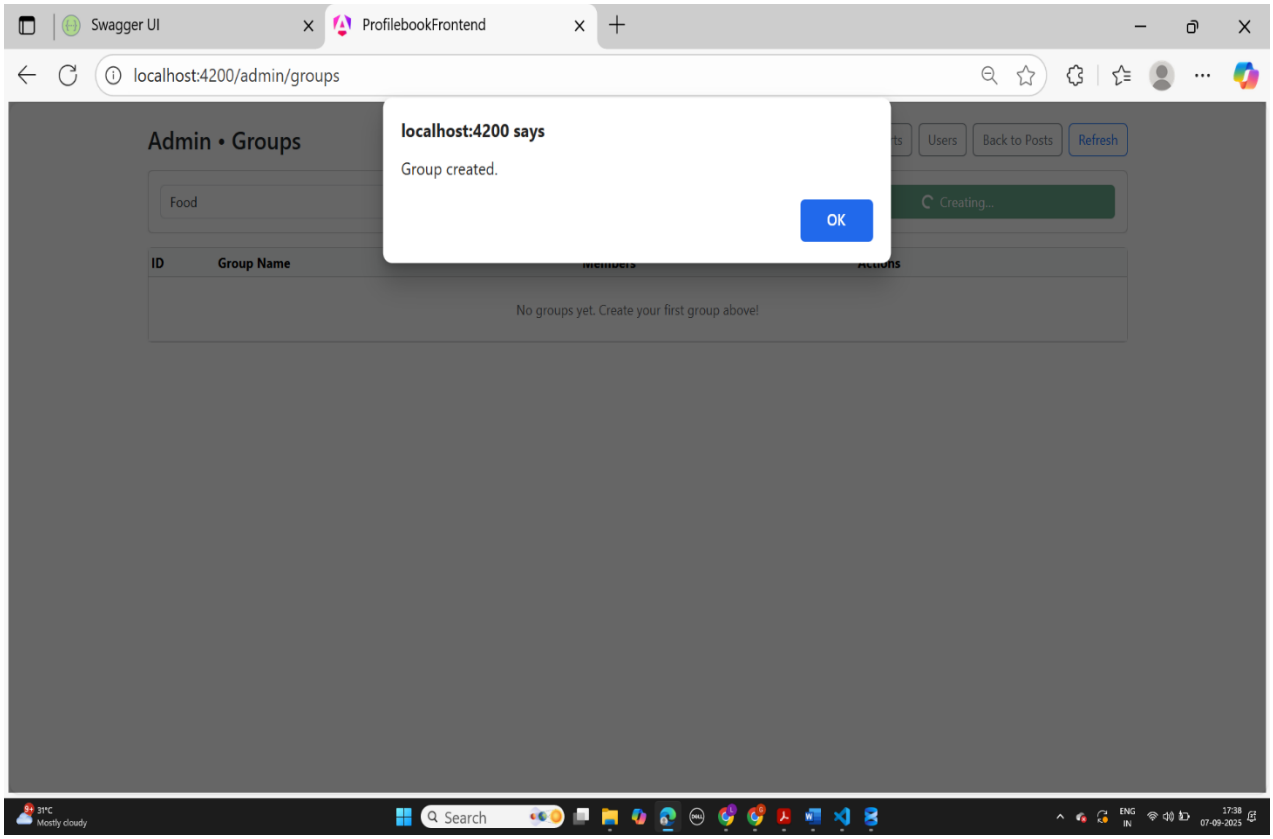
User Management Interface:

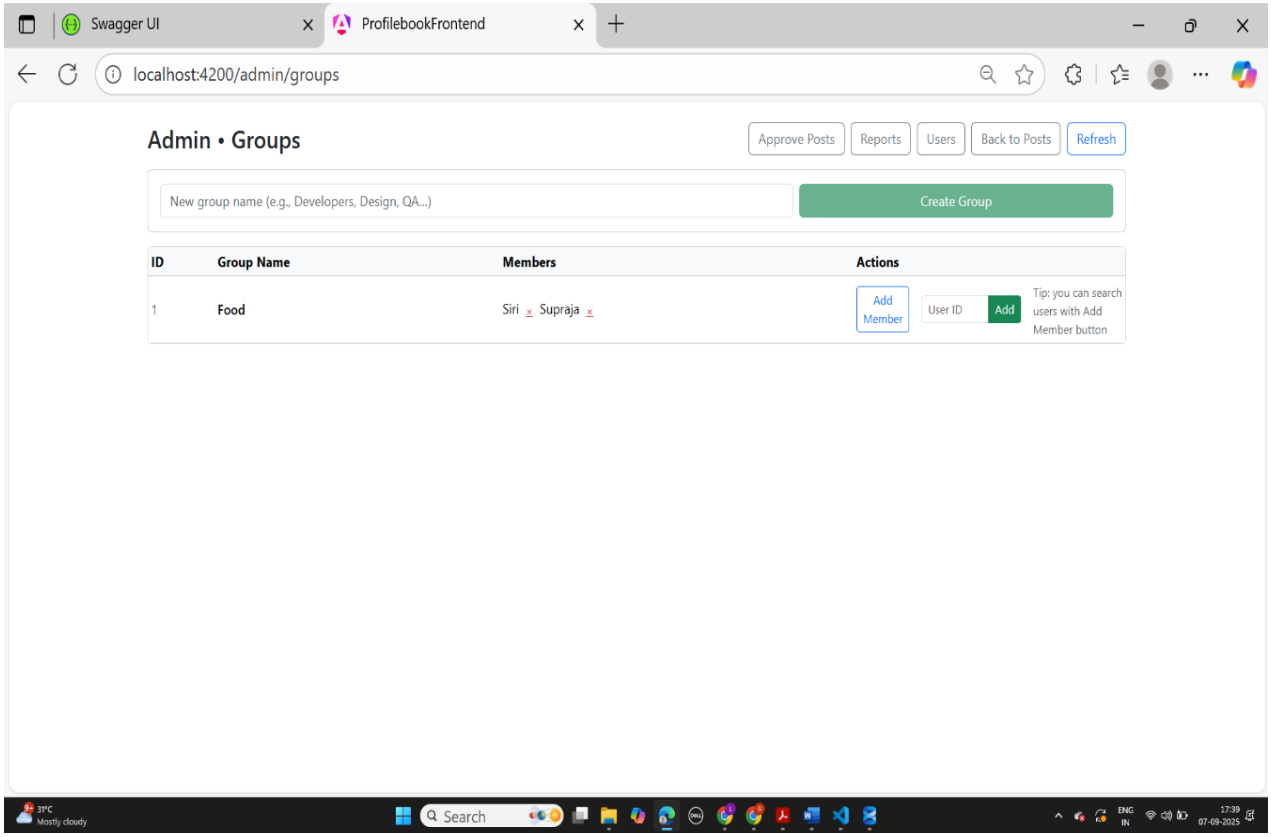


Reported Users List:



Admin managing groups




Swagger UI
ProfilebookFrontend
localhost:4200/admin/groups
Admin • Groups
Approve Posts
Reports
Users
Back to Posts
Refresh
New group name (e.g., Developers, Design, QA...)
Create Group
ID
Group Name
Members
Actions
1
Food
Siri
Supraja
Add Member
User ID
Add
Tip: you can search users with Add Member button
Search

CHAPTER 4: CONCLUSION

ProfileBook offers a comprehensive solution for online social interaction, combining ease of use, security, and functionality in a single platform. Future Scope: - AI-driven content moderation - Multi-language support - Push notifications - Enhanced analytics dashboard

References:

1. Microsoft ASP.NET Core Documentation
2. Angular Official Docs
3. SQL Server Documentation
5. Entity Framework Core Documentation